

BOZEMAN YELLOWSTONE, MT, USA

WMO: 726797

Lat: 45.788N

Lon: 111.161W

Elev: 4427

StdP: 12.49

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 24132

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-13.4	-6.5	-19.2	2.3	-12.6	-11.9	3.4	-5.9	24.4	36.8	20.4	35.6	3.1	180	0.611

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	34.5	92.0	61.2	88.6	60.3	85.3	59.4	64.3	83.4	62.6	81.9	61.1	80.5	7.9	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.0	84.9	69.4	55.8	78.1	68.2	54.1	73.4	67.1	31.6	83.6	30.3	81.8	29.1	80.5	86.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.6	18.2	15.3	DB	-23.4	97.3	7.0	3.6	-28.4	100.0	-32.5	102.1	-36.4	104.1	-41.4	106.8	(4)
(5)			WB	-23.7	69.7	6.8	5.6	-28.6	73.7	-32.5	77.0	-36.4	80.1	-41.3	84.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.3	21.6	24.1	34.2	42.4	51.2	59.1	68.2	66.0	56.5	43.6	30.6	20.8		
	DBStd	18.94	12.75	12.78	10.69	7.76	7.28	6.70	5.28	5.29	7.78	8.87	11.99	12.41		
	HDD50	4173	879	724	493	248	74	9	0	0	28	229	583	906		
	HDD65	8160	1344	1144	955	678	430	195	28	52	269	663	1031	1371		
	CDD50	1736	0	0	3	20	111	283	565	496	224	32	2	0		
	CDD65	247	0	0	0	0	2	19	129	82	15	0	0	0		
	CDH74	5339	0	0	0	11	130	547	2295	1775	554	27	0	0		
	CDH80	2171	0	0	0	0	22	182	1041	737	185	4	0	0		
(14) Wind	WSAvg	5.7	4.7	5.3	6.2	6.9	6.5	6.0	5.9	5.9	5.6	5.5	4.9	4.7		
(15) Precipitation	PrecAvg	14.5	0.6	0.5	0.9	1.3	2.4	2.4	1.1	1.2	1.4	1.1	0.7	0.6		
	PrecMax	20.0	1.7	1.5	2.4	3.2	6.2	7.3	4.6	3.3	3.1	3.1	2.3	1.3		
	PrecMin	8.7	0.1	0.1	0.3	0.2	0.4	0.3	0.1	0.0	0.0	0.0	0.2	0.1		
	PrecStd	2.9	0.4	0.4	0.4	0.7	1.1	1.3	1.0	0.7	0.9	0.7	0.5	0.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.4	55.4	67.7	75.4	83.1	91.3	98.1	95.8	90.4	77.6	64.7	51.5		
		MCWB	38.8	41.7	47.4	51.4	56.8	60.2	62.5	61.4	59.4	52.4	47.4	40.7		
	2%	DB	45.5	49.7	60.8	69.9	77.8	85.2	93.4	90.9	85.5	71.1	57.8	45.6		
		MCWB	37.2	39.0	44.3	49.5	55.3	60.7	61.3	60.8	57.2	51.1	43.8	37.5		
	5%	DB	41.8	44.6	55.8	64.0	72.9	80.3	89.7	87.7	80.7	65.3	52.4	41.7		
		MCWB	34.9	36.5	42.0	46.9	53.8	59.3	61.2	59.6	56.1	48.5	41.5	35.1		
	10%	DB	38.4	40.6	50.4	58.7	68.1	75.7	86.1	84.1	75.4	60.1	47.3	37.9		
		MCWB	32.9	33.9	39.5	44.5	52.0	57.4	60.4	58.8	54.4	46.3	38.9	32.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.0	42.9	48.3	53.6	59.4	65.0	68.3	68.1	62.5	55.6	48.7	41.3		
		MCDB	47.4	53.6	64.6	72.0	76.1	84.2	86.6	80.3	79.6	73.1	62.6	50.2		
	2%	WB	37.4	39.5	44.9	50.2	57.0	62.2	65.0	63.0	59.7	51.8	44.5	38.1		
		MCDB	44.1	48.2	58.8	67.1	73.6	80.6	84.3	83.0	79.2	67.7	55.9	44.8		
	5%	WB	35.3	37.0	42.5	47.7	54.9	60.2	63.4	61.2	57.4	49.6	42.0	35.5		
		MCDB	40.7	43.8	54.2	62.2	70.6	76.7	82.5	81.2	75.7	63.3	51.4	40.9		
	10%	WB	33.3	34.4	40.0	45.4	52.8	58.4	62.0	59.8	55.4	47.0	39.3	33.1		
		MCDB	38.0	39.9	49.8	57.3	66.3	73.5	81.6	79.8	73.3	58.6	46.9	37.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	19.6	20.8	23.2	24.8	25.9	27.8	34.5	34.8	31.4	26.4	21.9	19.9		
		MCDBR	21.6	23.9	31.3	35.9	35.7	37.3	40.0	41.0	41.0	36.7	28.6	20.5		
		MCWBR	14.9	15.2	17.0	18.2	16.9	16.8	14.5	15.7	17.3	19.0	16.8	14.0		
	5% WB	MCDBR	20.7	22.6	28.8	33.1	32.4	33.4	34.5	35.9	36.5	33.8	26.8	19.6		
		MCWBR	14.7	14.8	16.2	17.2	15.7	15.8	14.4	14.6	16.0	17.9	16.2	13.6		
(40) Clear-Sky Solar Irradiance	taub		0.244	0.253	0.273	0.310	0.320	0.329	0.336	0.344	0.320	0.284	0.258	0.241		
	taud		2.439	2.432	2.508	2.434	2.478	2.484	2.490	2.432	2.489	2.571	2.548	2.474		
	Ebn at Noon		282	300	308	302	300	297	293	287	286	285	276	272		
	Edh at Noon		21	26	29	34	34	34	33	34	29	23	19	18		
(44) All-Sky Solar Radiation	RadAvg		509	820	1192	1519	1740	2077	2241	1909	1429	893	544	403		
	RadStd		45	57	79	100	169	163	106	106	122	71	51	28		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	-1.15	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	-1.04	N/A	N/A	N/A	N/A

Nomenclature: See separate page